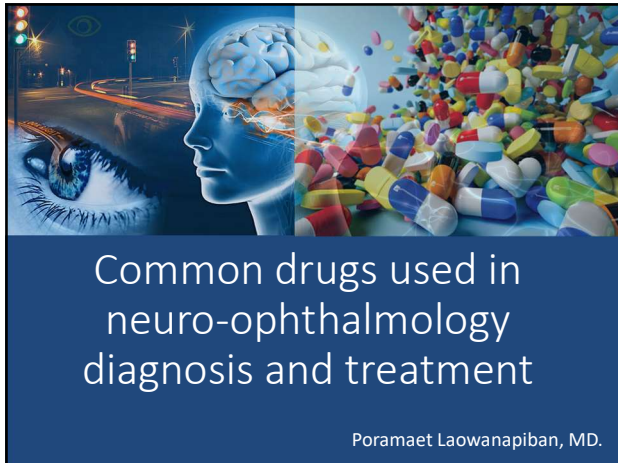




1

Neuro-ophthalmic drugs in diagnosis & treatment

Neuro-ophthalmic diseases	Drugs used for diagnostic/treatment
Optic neuropathies (Some), OMG	Corticosteroid, immunosuppression
Horner syndrome vs Physiologic anisocoria	Apraclonidine, Cocaine, hydroxyamphetamine
Adie pupil vs Partial CN 3 with pupil involvement	Dilute & non-dilute Pilocarpine
OMG vs other causes of ptosis & binocular diplopia	Pyridostigmine, neostigmine
IIH	Acetazolamide (Diamox)
Overactivity of CN 7 : BEB, HFS	Botulinum toxin A(Botox, Dysport)
Migraine	Sumatriptan, caffeine/ergotamine
LHON	Idebenone

2

Corticosteroids-mechanism of action

- Transactivation of anti-inflammatory genes : e.g., lipocortin inhibit PG and LT (anti-inflammation)
- Transrepression of proinflammatory genes : NF- κ B, IL-2 T cell proliferation, IL-2 B cell clonal expansion, Bcl-2 T-cell apoptosis in thymus (immunosuppression)
- Non-genomic effect : through various kinases e.g., reduce arachidonic acid creation

3

Steroid treatment of optic neuropathies

- Optic neuritis – Different causes with **variable urgency need and outcome of treatment**
- Ischemic and traumatic optic neuropathies: the initial injury is not inflammatory-Highly controversial
 - NAION-Controversial
- AAION (Giant Cell Arteritis)-Emergency and mainstay of Rx
- Compressive optic neuropathy, Malignant IIH, indirect TON – to **relieve the secondary inflammation** +/- before specific decompression treatment
- Non-infectious Infiltrative optic neuropathy : Sarcoidosis, Wegener's granulomatosis
- Infectious optic neuropathy : Controversial in conjunction with specific antimicrobial agents

REVIEW ARTICLE *Asia-Pac J Ophthalmology* 2018;7:218–228)
Steroid Treatment of Optic Neuropathies. Leanne Stunkel,* and Gregory P. Van Stavern

4

Giant cell arteritis - AAION

- Acute visual loss in one eye and highly suspicious for GCA: **immediately high dose systemic steroid (250 mg iv q 6 hr x 3 day)** to prevent the other eye from blindness
- Do not improve vision, but may prevent new ischemic events
- Other symptoms : Intermittent diplopia, Transient visual loss (with clinical of GCA, high ESR, CRP)
- Oral prednisolone (60-100 mg/day), with first dose immediately, maintenance with slow taper, with monitoring ESR, CRP and clinical over 1-2 yr (last 10 mg taper very slowly)

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5

Optic neuritis

- Acute treatment option in almost all causes of inflammatory optic neuritis
- ONTT regimen: 1 g iv OD or 250 mg iv q 6 hr for 3-5 days, 1 mg/kg/d for 11 days
- ONTT, 2015 Cochrane review : speed recovery but no significant difference of long-term visual outcome in MS-ON
- NMOSD :early treatment warrant better outcome, low threshold to plasma pheresis if refractory, recurrent
- ***Beware of infectious optic neuritis: required specific antimicrobial treatment

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Steroid Treatment of Optic Neuropathies. Leanne Stunkel,* and Gregory P. Van Stavern

6

Non-Arteritic Anterior Ischemic Optic Neuropathy

- limit Secondary damage from inflammation, swelling that further compression of capillaries in the optic nerve head
- Largest study (**Hayreh, 2008**): Prednisone 80 mg od for 2 wk
VA improve > 3 Snellen line in 70% treated and 40% control group, VF improve 40% in treated and 24.5% control group. **Lack of randomization & blinding, age, vascular risk not equal**
- 2 IV steroid studies(2014-retrospective , 2016-RCT): **no statistically significant difference**
- **Still controversial**, esp in lack of no proven treatment

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Steroid Treatment of Optic Neuropathies. Leanne Stunkel, Gregory P. Van Stavern

7

Traumatic optic neuropathy

- **Direct/Indirect TON** : Controversial**
- Delayed-onset damage due to inflammation, edema within confined optic canal, further compressed capillaries lead to ischemic damage
- **NASCIS II-1990**: (initial dose, 30 mg/kg, then 5.4 mg/kg/h for 23 hours) within 8 hours of spinal injury. **NASCIS III-1997**: extend to 48 hr (anti-oxidant and inhibit free-radical-induced damage, lipid peroxidase)
- **IONTS-1999** Non-RCT(57% observation, 52% steroid, 32% optic canal decompression improve > 3 lines of BCVA)
- **RCT-2013** high-dose IV steroid within 1 wk, show no significant different
- **CRASH study-2004**: steroid in traumatic brain injury result in significant more mortality (21% vs 17%) at 2 wk

REVIEW ARTICLE *Asia-Pac J Ophthalmology* 2018;7:218–228)
Steroid Treatment of Optic Neuropathies. Leanne Stunkel,* and Gregory P. Van Stavern

8

Steroids treatment in Ocular motility disorders

- Ocular myasthenia gravis
- Inflammatory orbital pseudotumor
- Tolosa-Hunt syndrome
- Trochleitis
- Ophthalmoplegic migraine
- Thyroid associated orbitopathy: active disease exacerbation, diplopia, compressive optic neuropathy (**OPTIC-2019** Trepotumumab, anti IGF-1R)

Gabrielle Weiner. Savvy Steroid Use. <https://www.aao.org/eyenet/article/savvy-steroid-use>

9

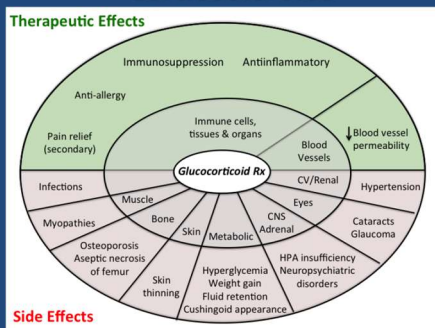
OMG Treatment

- A strategy of low-dose prednisone with gradual escalation appears to be safe, well-tolerated, and effective in treating OMG
- initiated at 10 mg every other day, and titrated to a maximum of 40 mg/day over 16 weeks, Treatment failure 100% vs 17% (**average required 15 mg/kg/d, mean time to minimal manifest = 14 wk**)

Efficacy of prednisone for the treatment of ocular myasthenia (EPITOME): A randomized, controlled trial. *Muscle Nerve*. 2016 Mar;53(3):363-9

10

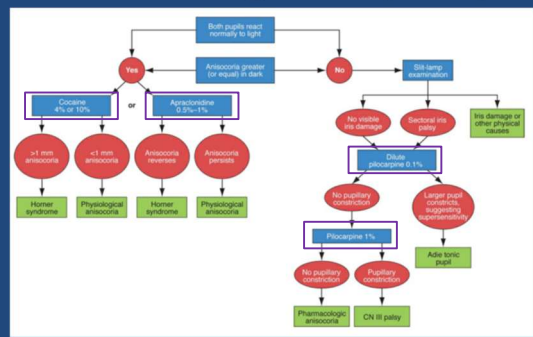
Therapeutic V.S. Side effects of corticosteroids



Adapted from Buttgereit et al, 2005

11

Anisocoria diagnostic eye drops



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12

Horner syndrome

- Small pupil abnormal (pupil dilator dysfunction)
- Anisocoria greater in dim light
- Dilatation lag of the miotic pupil in dark (anisocoria greatest at 5 sec and significantly less at 15 sec following dark exposure)
- No abnormal in pupil constriction to light or near
- Triad: **unilateral miosis, ptosis, anhidrosis**
- Iris heterochromia in congenital or rarely in long-standing

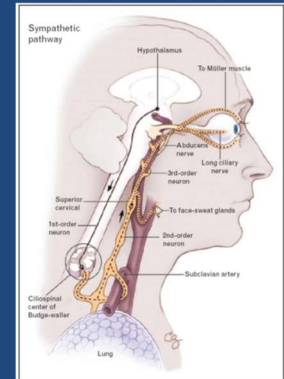
13

Sympathetic pathway to eye

1. Hypothalamus
2. Ciliospinal center of Budge
3. Superior cervical ganglion

- Pupil dilator
- Muller muscle + inferior tarsal ms
- Forehead/Face sweat fiber

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14

Horner syndrome

- Ipsilateral mild Ptosis (small pupil abnormal)



15

Pharmacologic testing of abnormal small pupil in anisocoria

• 4-10% Cocaine

Inhibit NE re-uptake by presynaptic nerve terminal

1 eyedrop BE at 0, 5 min and see result 45-60 min

Positive : Greater anisocoria of 0.8 mm or more



• 0.5-1% Apraclonidine

Weak α_1 -adrenergic agonist : denervation supersensitivity

1 eyedrop BE and see result at 30-45 min

Positive: Reversing anisocoria, ptotic eyelid will rise

Shortest report time : 36 h, variable negative 7 d

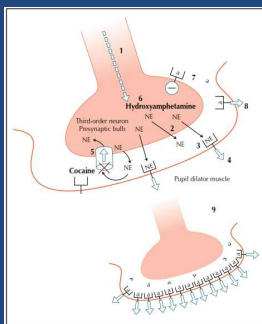
Contra in children under 2 y : CNS & respiratory depression

*If the test is negative in suspected Horner, perform Cocaine test on a separate day



16

Mechanism of action of pharmacologic eyedrops in Horner syndrome



17

4-10% Cocaine in Horner syndrome



- **Positive** : Greater anisocoria of 0.8 mm or more

A false-positive test result : synechiae

18



0.5-1% Apraclonidine in Horner syndrome



- **Positive:** Reversing anisocoria, ptotic eyelid will rise

19



1% hydroxyamphetamine



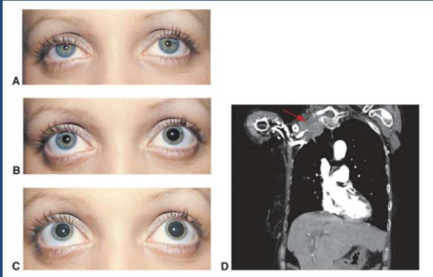
- **Help localization** : distinguish between a third order neuron and a first/second – order neuron lesion
- Cause release of NE into NMJ
- Must wait 48 h following cocaine or apraclonidine
 - **Bilateral pupil dilatation** : first and second-order neuron lesion
 - **Horner pupil fail to dilate** : third-order neuron / post-ganglionic lesion (**Greater anisocoria**)
- **High false positive and false negative rates**

20



1% hydroxyamphetamine

- RE ptosis & miosis in the light
- Anisocoria greater in the dark
- After 1% hydroxyamp hetamine ed



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21



Horner syndrome associated disorder

- **First-order Horner syndrome**: Wallenberg syndrome, tumor, cervical disc disease
- **Second-order Horner syndrome**: apical lung tumor (Pancoast syndrome), metastases, chest surgery, thoracic aortic aneurysm, trauma to brachial plexus
- **Third-order Horner syndrome**: degenerative changes in wall of carotid artery, vasospasm, ICA dissection, nasopharyngeal carcinoma extension into cavernous sinus

22



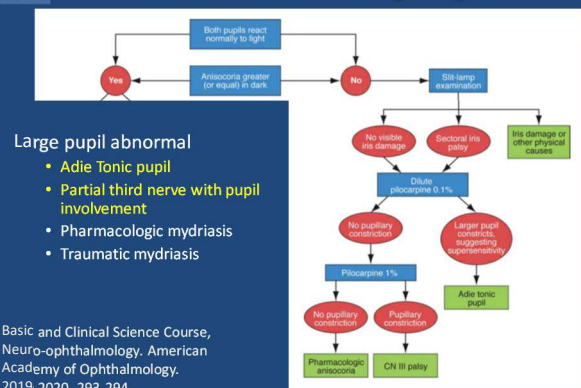
Horner syndrome diagnosis clues

- Careful history taking and examination are helpful in localizing Horner syndrome
- **First-order neuron**: numbness, weakness, ataxia, nystagmus
- **Second-order neuron**: trauma, cough, hemoptysis, swelling of neck
- **Third-order neuron**: numbness over CN V 1,2 and or 3 division, CN VI palsy
- **** painful Horner**, amaurosis fugax, altered taste (dysgeusia)

23

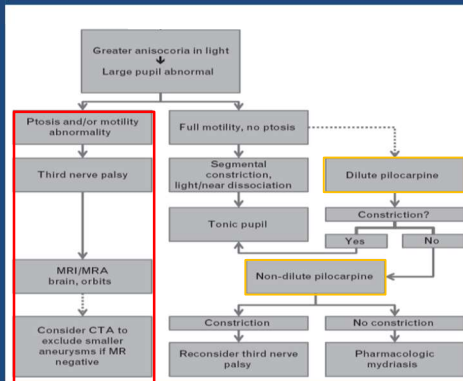


Anisocoria: Greater in bright light



24

Approach to large pupil abnormal



25

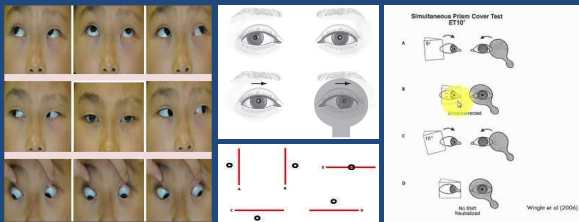
Partial third nerve palsy

Tips

- Detect subtle ipsilateral ptosis or SR, IO, IR, MR involvement (**simultaneous / prism cover, red glass test**)
- Confirm that ipsilateral ptosis is recently acquired
- *Old photo may help differentiate preexisting asymmetrical palpebral fissure
- The most concern of TNP with anisocoria (pupil involvement) : P-communicating artery aneurysm or compression
- MRI c MRA, CT c CTA or angiogram

26

Rule out subtle EOM involvement in partial CN 3 palsy



- Binocular blur/diplopia or no diplopia (poor VA or significant ptosis in any eyes)
- PCT, SPCT, single Maddox test

27

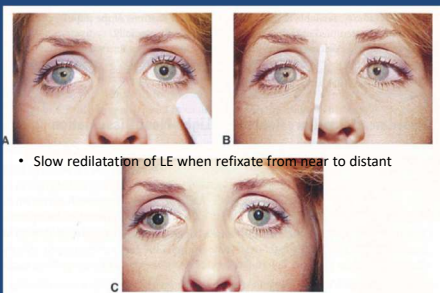
Tonic Adie pupil

- **Benign and idiopathic post-ganglionic parasymp fiber**
- Common: young women with or without visual symptoms
- Often sectoral or vermiform pupil constriction
- Light-near dissociation with slow, tonic movement
- Re-dilatation lag after accommodative constriction
- Miosis in old Adie

28

Light-near dissociation in Adie's tonic pupil = aberrant regeneration of post-gg parasymp

- Strong consensual light response on RE
- Both pupil constrict to near response



- Slow redilatation of LE when refixate from near to distant

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29

Tonic Adie pupil

- Over a period of years, become miotic
- Other DDx: autonomic disorders, ischemia, or any orbital process
- **Usually benign, not required imaging**
- **MRI when suspicious of orbital process, systemic process in bilateral cases
- **Holmes-Adie syndrome** : diminished deep tendon reflex, orthostatic hypotension

30

Pharmacological tests in abnormal dilated pupil

- After exclude trauma, history of possible exposure to parasympatholytic or sympathomimetic agents
- Pilocarpine: muscarinic agonist act at NMJ
 - Dilute (0.05-0.15%) pilocarpine
 - Nondilute (1-2%) pilocarpine
- Tonic pupil (> 2 wk): upregulate receptor → hypersensitive to dilute pilocarpine



31

Denervation supersensitivity in Adie's pupil



Figure 10-7 A, Right Adie tonic pupil. B, After instillation of pilocarpine (0.1%), the right pupil becomes miotic, demonstrating supersensitivity. (Courtesy of Lanning B. Kline, MD.)

- Present in 80% of Adie's

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32

Drugs in OMG diagnosis & treatment

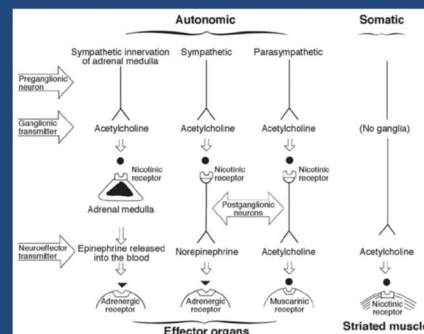
Indirect-acting cholinergic agonist: Ach esterase inhibitor, increase the level of Ach thus stimulate both Nicotinic and Muscarinic Ach-R

- Neostigmine(Prostigmine®)
- Edrophonium (Tensilon®)
- Pyridostigmine (Mestinon®)



33

Cholinergic receptors



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34

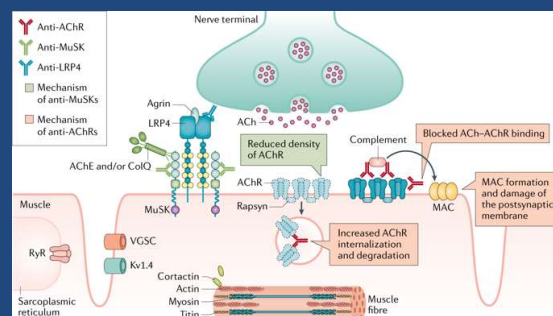
OMG diagnosis & treatment

- OMG remains a clinical diagnosis
- Pharmacologic, serologic and electrophysiologic testing can **help in increasing the diagnostic accuracy**
- Therapy should aim at **achieving satisfactory symptomatic control while reducing minimizing the treatment-induced side effects/toxicities.**

Curr Opin Ophthalmol. 2018 Nov;29(6):477-484. **Ocular myasthenia gravis: an update on diagnosis and treatment.** Fortin E, Cestari DM, Weinberg DH

35

MG autoimmune disease



Anti-Ach R, anti-MuSK, anti-LRP-4

36

Tests for myasthenia gravis

Table 2
Tests for myasthenia gravis

Test Name	Positive Result	Statistics
Sleep or rest test	Improving symptoms after a period of rest	Moderate sensitivity and specificity
Ice test	Improvement of ptosis after icepack is applied to the eyelids for several minutes	Moderately sensitive and high specificity
Fatigue test	Eyelid fatigue with upward gaze	High specificity
Serologic test for anti-AChR antibodies	Positive for anti-AChR antibodies (binding, blocking, and modulating forms)	Moderate sensitivity but high specificity
Repetitive nerve stimulation	Decrement in compound muscle action potential of stimulated nerves	Low sensitivity and high specificity
Single fiber EMG	Abnormal jitter of weak muscles	High sensitivity but may not be universally available
Edrophonium test	Symptom improvement with edrophonium	High sensitivity and specificity but largely supplanted by less invasive clinical and laboratory tests discussed

37

Edrophonium (Tensilon®) test

- Intravenous
- Onset within 30-60 sec, duration 5-10 min
- 0.10 mg/kg in children, 0.05-1.0 mg sc in infant
- < 10 mg in adult via 2,4,4 mg increment q 2 min
- Worse EOM due to depolarizing block by excess of Ach
- Good with ptosis

38

Prostigmine (neostigmine®) test

- Intramuscular
- Onset 15 min, peak 30 min, duration 45-several hr
- 1.0-1.5 mg in adult
- Suited for detailed and prolonged exam of ocular motility, diplopia test
- **Children:** 0.025 to 0.04 mg/kg I.M. preceded by 0.011 mg/kg S.C. of atropine sulfate.
- **Adults:** 0.022 mg/kg I.M. Give atropine 0.011 mg/kg I.V. with dose or I.M. 30 minutes before dose. If cholinergic reaction occurs, stop test and give atropine sulfate 0.4 to 0.6 mg I.V.

39

Acetylcholine esterase inhibitor in OMG treatment

- Pyridostigmine : first-line for **symptomatic relief**
- Started at dose of 30 mg tid up to 150 mg qid
- Pediatric dose (5–7 mg/kg/day, 3 times/day)
- Ptosis may improve the most, diplopia may have little to no response
- Anti-MuSK-positive : less responsive

- Fail to cause remission of disease
- Not change the progression rate to generalized MG

40

Cholinergic side effect

- slow heart beat, possibly leading to cardiac arrest
- muscle weakness, muscle cramps, and muscle pain
- convulsions
- weak breathing, inability to breath
- increased stomach acid and saliva

41

Muscarinic side effect of cholinergic agonist drugs

- | | |
|--|---|
| • S - Salivation | D - Diaphoresis and Diarrhea |
| • L - Lacrimation | U -Urinary frequency |
| • U -Urinary frequency | M -Miosis |
| • D -Diarrhea | B -Bronchospasm and Bronchorrhea |
| • G -Gastrointestinal cramping and pain | E – Emesis |
| • E - Emesis | L – Lacrimation |
| • M - Miosis | S – Salivation |

42

Nicotinic Cholinergic agonist side effect

- Muscular weakness
- Muscular fatigue and fasciculation
- Respiratory muscle weakness
- Tachycardia
- Hypertension

43

Ophthalmic indication of Botulinum toxin-A



- Hemifacial spasm : AICA, CP angle tumor, post-paralytic (Bell's palsy)
- Blepharospasm with lid opening apraxia : Parkinson's disease, PSP, isolated loss of levator muscle control, blepharospastic apraxia
- BEB + Oromandibular dystonia in Meige syndrome
- Lid retraction in TED
- Strabismus: CN 6 palsy-ischemia, post trauma
- Eyelid myokymia
- Aberrant regeneration of eyelid protractor, crocodile tear syndrome
- Spastic entropion
- Headache syndrome
- Hyperhidrosis and Lacrimal Hypersecretion Syndromes

44

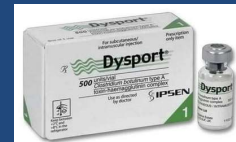
Botulinum toxin-mechanism of action

- bind to high-affinity sites on the presynaptic cholinergic nerve terminals
- decrease release of Ach, cause a neuromuscular blocking effect
- BoNT-A and BoNT-E cleave **synaptosome-associated protein (SNAP-25)**, a presynaptic membrane protein required for fusion of neurotransmitter-containing vesicles

<https://www.youtube.com/watch?v=bFUeaCsTems&t=2s>

45

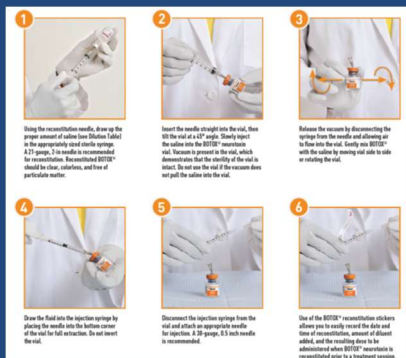
Botulinum toxin



- Botox 100 u/vial reconstituting with sterile saline 4-2 ml (2.5-5 unit/0.1 ml)
- Dysport 500 u/vial + NSS to 2.5 ml (20 unit/0.1 ml)

46

Botox preparation



47

Botulinum toxin : Useful facts

- Preparation: not to foam the solution
- Manufacturer recommend using solution within 4 hr
- Should not be frozen after preparation
- Must be stored at 2-8 C and used within 24 hr
- Some studies report safely used for up to 2 wk without noticeable decrease in clinical efficacy
- begins to take effect within 24-72 hours and reaches maximum effect within 7-14 days

48

Acetazolamide (Diamox®)



Acetazolamide mechanism of action

- A carbonic anhydrase inhibitor
- accumulation of carbonic acid : reversible reaction into bicarbonate and a hydrogen ion
- Carbonic anhydrase is found in RBC and proximal tubule of the kidney
- Na K and HCO_3^- get excreted, also excretion of excess water
- The clinical result is a decrease in blood pressure, **decreased intracranial pressure**, and decreased intraocular pressure
- Bicarbonate excretion also increases the acidity of the blood. Aqueous humor levels decrease in the eyes, compensatory hyperventilation

49

50

Contraindication of acetazolamide

- Impaired liver and kidney function
- Hypo K, Na, Hyper Cl,
- Prolong QT
- ****Allergy to sulfonamides**
- Drug interaction : reduced amphetamine clearance, increase excretion of lithium, decrease excretion of phenytoin, primidone, quinidine
- Anti-folate drugs e.g., methotrexate and trimethoprim
- Cat-C for pregnancy, should only be used if the benefits outweigh the risks

Acetazolamide in IIH treatment

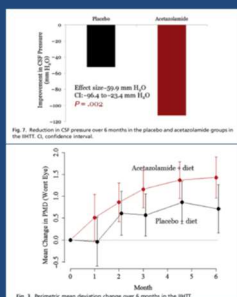
- Initial dose 0.5-1 g/d
- Many start at 250 mg bid (some consider too low)
- Recommended 500 mg bid (250 mg qid)
- Max dose 3-4 g/d (most cannot tolerate, 56%)
- Side effect include extremity paresthesias, fatigue, metallic taste from carbonated beverages, decreased libido

Headache. 2017 Sep;57(8):1303-1310. doi: 10.1111/head.13144. Epub 2017 Jul 30.
The Idiopathic Intracranial Hypertension Treatment Trial: A Review of the Outcomes.
Smith SV¹, Friedman DI^{1,2}.

51

52

Significant Acetazolamide effect, IIHTT



- Decreased CSF pressure
- Decrease papilledema grade
- Improve visual field function
- Improve quality of life measures

Update on Idiopathic Intracranial Hypertension
Neurologic Clinics, Volume 35, Issue 1,
February 2017, Pages 45-57
Michael Wall

53

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- Basic and Clinical Science Course, Neuro-ophthalmology. American Academy of Ophthalmology. 2019-2020
- Gabrielle Weiner. Savvy Steroid Use.
<https://www.aao.org/eye/article/savvy-steroid-use>

54